Quantitative Political Science Research Is Greatly Underpowered

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The social sciences face a replicability crisis. A key determinant of replication success is statistical power. We assess the power of political science research by collating over 16,000 hypothesis tests from about 2,000 articles in 46 areas of the discipline. Under generous assumptions, we show that quantitative research in political science is greatly underpowered: the median analysis has about 10% power, and only about 1 in 10 tests have at least 80% power to detect the consensus effects reported in the literature. We also find substantial heterogeneity in tests across research areas, with some being characterized by high power but most having very low power. To contextualize our findings, we survey political methodologists to assess their expectations about power levels. Most methodologists greatly overestimate the statistical power of political science research.

Statistical power is critical to any discipline that practices null hypothesis significance testing, such as political science. Power—or the probability that a statistical test will reject the null hypothesis when the alternative is true—must be a key consideration when researchers design a study and when readers evaluate the credibility of published findings.

As power increases, the probability of committing a false negative error (type II) goes down. When power is low, empirical findings are less likely to replicate (Altmejd et al. 2019). When power is low and one happens to find a statistically significant estimate, that estimate is often much greater than the "true" underlying effect, and it may well have the wrong sign (Gelman and Carlin 2014; Gelman and Tuerlinckx 2000; Ioannidis, Stanley, and Doucouliagos 2017). As political scientists grapple with the replication crisis (Baker 2016), it is crucial to know whether our research designs have sufficient power to generate credible findings.

The power of a study is affected by a number of factors including the desired level of statistical significance, sample size, measurement variability, and the magnitude of the effects under investigation. Typically, larger effect sizes and larger samples result in higher power.[1] As such, it seems reasonable to expect that power will vary from study to study and across scientific fields and disciplines.

In this article, we examine statistical power in political science by assembling a dataset of 16,649 hypothesis tests, which are aggregated to produce 351 meta-analytic estimates,

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1. Power calculations are conducted based on assumptions regarding effect and sample size but also by considering other factors such as measurement error. Design-based inference provides researchers with some control over the number of observations collected, yet sample size is often driven primarily by budgetary concerns rather than power calculations.

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reported in 46 peer-reviewed meta-analytic articles.[2] We estimate power retrospectively by leveraging estimates of mean population effects from the meta-analyses.[3] In essence, we calculate the power of each test to detect the consensus effect reported in its literature. Our results suggest that quantitative political science research is greatly underpowered. The median research result has about 10% power to detect this effect ($\alpha = .05$), and only about 1 in 10 statistical tests have at least 80% power.[4] As we show below, there are some indications that power may be increasing gradually over time, but our estimates remain low, even in the later parts of our sample.

These results are both dispiriting and surprising. To contextualize them, we conducted an expert survey of political methodologists to measure their expectations about power levels in published hypothesis tests. On average, the experts in our sample believe that 66% of studies have at least 50% power and 43% have at least 80% power. We demonstrate that these expectations are overly optimistic. On average, experts appear to overestimate the share of studies powered at the 50% level by 48 percentage points and the share of studies powered at the 80% level by 32 percentage points. Political science research suffers from low power, and this problem should be taken seriously by empiricists and methodologists.

The rest of this article is structured as follows. We begin by describing our research design and data, and by arguing that meta-analytic estimates of population mean effects for various research questions present the best opportunity for estimating power retrospectively. Then, we explain how we surmount challenges related to publication bias, review the data we collected from meta-analysis replication files, and introduce our expert survey. Next, we propose various extensions and robustness checks to our analyses. Finally, we conclude with a discussion of low power, and propose a number of institutional, methodological, and theoretical remedies to this challenge.

METHOD AND DATA

Our goal is to assess statistical power in recent quantitative political science research. In other words, we want to estimate the probability that any given statistical test will reject the null hypothesis, for a given "true" effect size. To achieve this, the ideal dataset would have a standard error along with a "true" effect size for every estimate reported in peer-reviewed journals in political science and closely adjacent fields. Given these data, one could calculate retrospective power for each test at some α level such as .05.[5] A test for which the standard error is less than the "true" effect size divided by 2.8 would have at least 80% power.[6]

Of course, in general "true" effect sizes are unknown. One can respond to this challenge in a number of ways. The most problematic approach is to use the reported effect size of each test to calculate the power of that test. This circular form of post hoc power calculation is rightly shunned as it reveals no new information and instead merely recapitulates *p* values (Hoenig and Heisey 2001). Perhaps worse, when there is selection on statistical significance, such an approach will tend to dramatically overstate power (Gelman 2019).

A second approach is to judge power against theoretically informed minimum effect sizes of interest. One could do this for an entire body of literature if the articles in it typically reported their minimum effect size of interest, but this is uncommon in political science. If articles report standardized effect sizes, then one can use a related approach of measuring power against arbitrary but equally sized "small" or "medium" effect sizes. However, again political scientists rarely report standardized effect sizes, and so again this approach is not practical at the scale of our broad survey.

We therefore use a third approach: estimating power retrospectively using meta-analytic estimates of population mean effects for various research questions. This follows previous work in neuroscience (Button et al. 2013), economics (Ioannidis et al. 2017), psychology (Stanley, Carter, and Doucouliagos 2018), and ecology and evolutionary biology (Yang et al. 2022). Because meta-analyses encompass all relevant, reported hypothesis tests for specific research areas, meta-analysis provides the best and most widely informed estimate of the population mean effect. Perhaps the only exception are estimates from preregistered multilaboratory replications (Klein et al. 2018; Open Science Collaboration 2015),

2. We define a "meta-analysis" as a grouping of at least five comparable estimates that researchers have aggregated to calculate meta-analytic effects. A single meta-analytic article often reports many meta-analyses. Some meta-analyses address closely related substantive questions using slight variations on the independent and dependent variables, or using different sets of estimates (e.g., experimental vs. observational).

3. This should not be confused with post hoc power analysis, which we discuss and strongly discourage in the "Method and Data" section.

4. One potential concern is that our sample includes estimates from meta-analyses regardless of whether said meta-analysis rejected the null hypothesis. Meta-analyses in which the null was not rejected might have near-zero population mean effect estimates, and this could drive down our overall estimates of power. However, we find the median level of power in the subset of studies in which the meta-analytic estimate is itself significant at .05 level is only 0.9 percentage points higher.

5. Throughout the article, we use an α level of .05.

6. Put differently, to discriminate a "true effect" from zero with 5% significance and 80% power, the effect's standard error needs to be smaller than the absolute value of the underlying effect divided by 2.8. This relationship is derived from the standard normal value that makes a 20%/80% split in its cumulative distribution and the usual value of 1.96 for a significance level of 5% (Cohen and Wolman 1965).

but these are uncommon in political science. Using meta-analyses allows us to measure the power of each test against the relevant literature's most informed estimate of its effect size.

Meta-analyses

To find meta-analyses, we searched through the archives of 141 peer-reviewed journals in political science and closely adjacent fields. The list of peer-reviewed journals in the scope of our data collection came from two sources. First, we selected all journals appearing in the social science subcategories "Political Science," "Diplomacy & International Relations," and "Public Policy & Administration" of Google Scholar Metrics' top publications as of 2021. Second, we selected the 50 journals with the highest total citations for the year 2020 in the categories "political science," "international relations," and "public administration" categories of Clarivate's Journal Citation Reports. The full list of journals is printed in appendix A. We conducted full-text searches for the keyword "meta" in the archives of all these journals, ignoring all results dating from before 2000.[7] All keyword matches were checked manually to retain articles for which authors gathered data from other articles in order to run a quantitative meta-analysis.

We excluded qualitative meta-syntheses and articles in which authors combine multiple estimates from their own work, using meta-analytic techniques as a form of model averaging. We also excluded articles not focused on political topics, that is, articles in which both the outcome and explanator variables are primarily the object of research in other disciplines, such as economics, criminology, management, and psychology. Ambiguous cases were reviewed by two different coauthors of the present article. In total, 85 articles met our definition of a meta-analysis in political science. We were able to obtain the data from 46 of those meta-analytic articles from public journal archives, research repositories, and author websites, by transcribing results from published tables and figures, and by contacting authors directly.

One potential concern is that because we were able to obtain data for only 46 out of 85 eligible meta-analyses, we cannot definitively reject the possibility that nonpublic data would differ in terms of power or research area within political science. However, the decision of meta-analysts to make their data publicly available is likely unrelated to the power of the individual studies' estimates, because those studies were rarely conducted by the meta-analysts themselves. Consequently, although we acknowledge this limitation, it seems unlikely that this would significantly affect the overall conclusions of our analysis. Indeed, in appendix A, we present a balance table comparing the 46 articles in our sample to the 39 articles not in our sample. This table suggests that there are no meaningful differences between both sets of meta-analyses in terms of subfield distribution, the proportion looking at a research question related to political behavior, and the proportion published in a "top three" political science journal (i.e., *The Journal of Politics*, *American Political Science Review*, and *American Journal of Political Science*). We do observe a slight difference in terms of the average publication date: articles in our sample were published in 2017 on average, versus 2014 for articles not in our sample. This may be due to replication archives becoming more common over time (Rainey et al. 2024).

Population mean effects

The main drawback of using meta-analyses for retrospective power analysis is that estimates of the population mean effects are based on reported results, which may have been selected based on statistical significance. Examples of such selection include the file drawer problem, reporting bias, specification searching, *p*-hacking, and the garden of forking paths.[8] Selection on statistical significance will result in there being too few statistically nonsignificant estimates and too many statistically significant estimates, which will inflate the size and significance of the collection of estimates reported in the literature.[9] If meta-analyses aggregate inflated estimates, they will likely produce inflated estimates of population mean effects, which will in turn lead us to overstate power (Gelman 2019; Kvarven, Strømland, and Johannesson 2020).

There is no universal agreement in the methodological literature on a best approach for estimating population mean effects from reported results. Thus, we use three alternative methods, which we introduce from least to most aggressive in how they attempt to correct for publication bias.

The first method is the unrestricted weighted least squares (UWLS). The UWLS is a simple weighted average of the form $\hat{\mu}_w = \sum(1/\phi\sigma_i^2)y_i / \sum(1/\phi\sigma_i^2)$, where y_i are study-level

7. The year 2000 cutoff point is arbitrary and was chosen due to resource constraints and data availability. It is useful to note that data collection involved conducting full text searches on the full universe of articles published by a large sample of peer-reviewed journals (listed in the following subsection). Through this process we found that very few meta-analyses had been published in political science before 2000 and that the data for these articles were generally unavailable. The median publication year of the meta-analyses in our sample is 2010.

8. The presence of publication bias is "one of the strongest findings across the sciences" (Berinsky, Druckman, and Yamamoto 2021, 370).

9. As noted in *The Journal of Politics*' Pre-Registration Guidelines, "if power analyses and smallest effect sizes of interest are based on effect sizes reported in previous studies, authors should keep in mind that meta-scientific studies reliably report inflated effect sizes in the reported literature that often shrink in replication studies" (*The Journal of Politics* 2022).

estimates, σ_i^2 are within-study variances, and ϕ is a scaling factor. UWLS can be estimated by regressing the study-level estimates on a constant using weighted least squares with weights equal to $1/\sigma_i^2$.[10]

The second method is the weighted average of the adequately powered (WAAP), introduced by Stanley, Doucouliagos, and Ioannidis (2017). We compute the WAAP in three steps: (1) calculate the UWLS; (2) use the UWLS to estimate the power of each study; (3) calculate the UWLS again, but using only the subset of estimates that exceed 80% power.

The final method is our most aggressive strategy to unwind publication bias: the precision-effect test and precision-effect estimate with standard error (PET-PEESE) (Egger et al. 1997; Stanley 2017; Stanley and Doucouliagos 2014). The intuition that motivates this approach is that selection on statistical significance will result in a positive relationship between reported effect estimates and reported standard errors. One can thus regress estimates on standard errors or standard errors squared, interpreting the intercept of this regression as the estimated effect when the standard error is equal to zero.[11] In a psychology study comparing 15 meta-analyses to multilaboratory replications on the same research questions, PET-PEESE produced the least inflated estimates of the four methods tested (Kvarven et al. 2020).

Each of these three methods produces alternative estimates of population mean effects for each meta-analysis in our sample. With these in hand, we then calculate the statistical power of each estimate based on its standard error, one of our estimates of the population mean effect from the relevant meta-analysis, and an α level of .05. As we show below, results obtained using the three techniques are broadly consistent. For simplicity, the rest of this article generally focuses on results obtained via UWLS, the most "generous" approach.[12]

Expert survey

We conducted an original expert survey of political methodologists in June 2022 to assess their expectations about power levels.[13] Our sample includes all authors who published in the peer-reviewed methodology journal *Political Analysis* between 2010 and 2021. In total, 131 methodologists answered our survey, for a response rate of 27%.[14]

We asked experts to consider all hypothesis tests published in the 50 peer-reviewed journals with the highest impact factors in political science and closely adjacent fields over the past two decades. They then guessed what share of those tests had at least 50% and 80% power to reject the null at the $\alpha = .05$ level. To align the survey question with our empirical approach, we asked about the share of tests that had at least 50% or 80% "power to reject the null hypothesis." This guided respondents to think about power relative to likely effect sizes. For logical consistency, we exclude a few respondents who stated that more studies have at least 80% than 50% power. Forty-two respondents said that they did not know the answer to at least one of the questions.

RESULTS

We begin by examining selection on statistical significance. Figure 1 shows a histogram of the absolute value of z-statistics for all hypothesis tests.[15] The distribution is right-skewed: there are many significant tests. The large spikes and humps show that many z-statistics are concentrated at or just above two conventional thresholds of statistical significance: 95% and 99%. These results resemble those found by Gerber and Malhotra (2008) for political science and by Brodeur et al. (2016); Brodeur, Cook, and Heyes (2020); and Gorajek and Malin (2021) for economics.[16] It is difficult to explain such a distribution of z-statistics in the absence of selection on statistical significance.

10. UWLS produces the exact same point estimates as the more common fixed effects meta-analysis, but simulations suggest that the UWLS standard errors have better properties (Stanley and Doucouliagos 2024). Because our retrospective power calculations only require meta-analytic estimates of the population mean effects and not their standard errors, the two approaches are functionally equivalent for our purposes.

11. Following Stanley and Doucouliagos (2014), we first regress the effect estimates on their standard errors and a constant. If the constant from this regression is not significant at $p < .1$, we use it as our estimate of the population mean effect. If the constant is significant at $p < .1$, we regress the effect estimates on the standard errors squared and a constant, using the constant from this regression as our estimate. This conditional approach is used because the linear model is better in the absence of a "true" effect, whereas the squared model is better in the presence of a "true" effect. All regressions in the PET-PEESE use weighted least squares, where the weights are equal to $1/\sigma^2$.

12. We focus on the UWLS estimates because our core finding is that statistical power is generally lower than expected by political methodologists, and we want to avoid "stacking the deck" in the direction of this conclusion by using an aggressive bias correction strategy.

13. Our analysis of the survey data was preregistered following the AsPredicted template, and the preanalysis plan is available at https://aspredicted.org/blind.php?x=G28_YCP. The design was approved by two different institutional review boards from the universities of two coauthors of the present article. See app. C for our recruitment materials and survey instrument.

14. We emailed 478 methodologists (555 minus 77 for whom emails were undeliverable). In total, 131 answered all of our questions, yielding a response rate of 27%.

15. We lack information on degrees of freedom for most observations, so we calculate z-statistics instead of t-statistics. For visual clarity, we exclude z-statistics larger than 10.

16. See especially fig. 1 by Brodeur et al. (2016). Vivalt (2019) shows similar results for quasiexperimental impact evaluations.

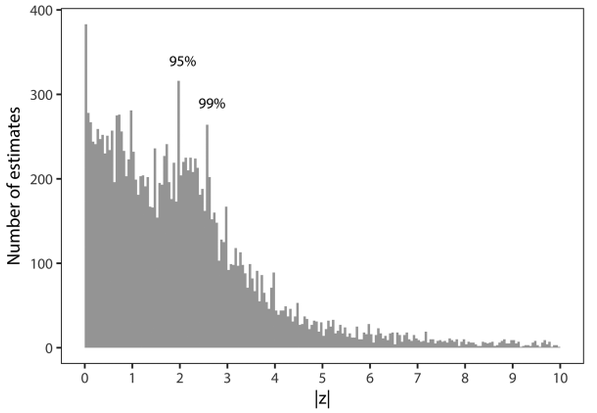

Figure 1. Distribution of *z*-statistics in the full sample of estimates. The dashes highlight the conventional thresholds of statistical significance of 95% and 99%.

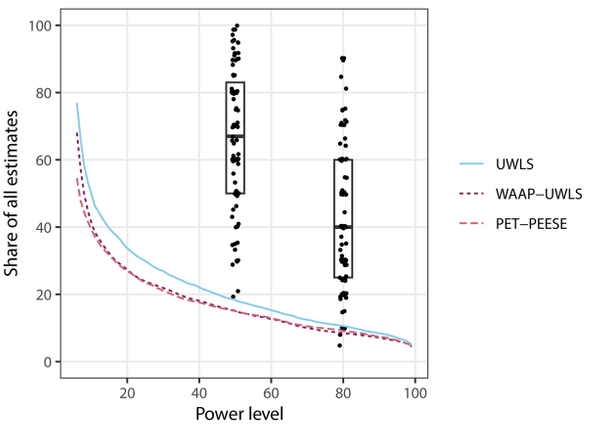

Figure 2. Retrospective power analyses and the view from political methodology. Lines represent the share of estimates by power level, using three different approaches to estimate the population mean effect. Black dots and boxplots represent the distribution of responses in the expert survey.

Figure 1 has important implications for our study of statistical power. Because reported findings are characterized by publication bias, our estimates of the "truth" and our power calculations are likely to be inflated. The UWLS results presented below should thus be interpreted as a best-case scenario for statistical power in the discipline.

Our key results are presented in figure 2. The results for each of the three methods for estimating population effects are displayed using differently line types, which represent the share of all hypothesis tests that reach a given power level from 6% to 99%.[17]

All three techniques produce similarly shaped curves. UWLS, which addresses publication bias least aggressively, yields the highest power estimates. Yet power is low even under these generous conditions: only half of the tests reach 10% power, a fifth reach 50% power, and a tenth reach 80%. Our two other methods for estimating population mean effects result in even lower power levels.

The black overlays in figure 2 display the results of our expert survey. As a reminder, we asked respondents to estimate the share of hypothesis tests that achieve at least 50% power and at least 80% power within all articles published since 2000 in the top journals in political science and related fields. Their responses are shown in black dots, and the boxplots mark the quartiles for each distribution. The curves for our power calculations barely overlap with the expert responses: methodologists appear to overestimate the power of hypothesis tests by a large margin.

The gap between expert assessment and estimated power could stem from several factors. First, methodologists could suffer from recency bias, as power appears to have increased slowly over time (see fig. 6). Second, the studies in our sample may not match the studies, journals, or subfields that survey respondents had in mind when contemplating our question. Finally, there remains considerable uncertainty in our own estimates of power. Taken together, these factors may explain part—but probably not all—of the apparent overoptimism displayed by political methodologists.

EXTENSIONS AND ROBUSTNESS

We conducted several complementary analyses and robustness checks. In the online appendix, we report alternative tests, which confirm that our main findings are not driven by outliers. Here, we report results that refine the interpretation and add nuance to our findings.

Minimum effects of interest

For this article we calculated power retrospectively based on meta-analytic estimates of population mean effects. In practice, however, experimentalists in political science often design their studies based on prospective power calculations anchored by the size of a "minimum effect of interest" (MEI). Our results can tell us if studies are sufficiently powered to detect consensus effects in the literature; they do not directly tell us if studies are well powered to detect the MEIs targeted by individual researchers. But even if we cannot directly characterize power with respect to MEIs, our results suggest useful informal bounds. Consider two cases.

First, if MEIs are systematically smaller than population mean effects, our analyses would overstate the power of political science. In that case, our results could be interpreted as an upper bound for the average power of the studies in our

17. The share is the proportion of all reported tests that have power at least as high as displayed on the horizontal axis.

sample; the problems that we highlight in this article would be even more worrying. Second, if MEIs are systematically larger than meta-analytic estimates, our analyses would understate the level of power in political science research.[18]

In figure 3 we probe the sensitivity of our conclusions to this potential issue by artificially inflating all population mean effect estimates by a factor of five and replicating our analyses.[19] After this five-fold inflation, the power estimates align more closely with the expert survey responses, which gives an approximate sense of the degree of overestimation of power in the expert community.

Substantively, we find that one-third of tests fail to reach 50% power even after the inflation. In other words, even if researchers were designing studies to target effect sizes five times larger than the consensus estimates reported in the literature, quantitative political science research would still be greatly underpowered. Another way to circumvent the problem posed by the disconnect between population mean effects and MEIs is to shift the focus away from power, toward the magnitude and sign of point estimates.

Magnitude and sign

Low-powered studies that are subsequently filtered for statistical significance are more likely to report effects that are inflated or of the incorrect sign (Gelman and Carlin 2014; Gelman and Tuerlinckx 2000; Ioannidis et al. 2017). We test for this in our data, essentially offering empirical analogues to the type M and type S error rates described in Gelman and Carlin (2014).

Among our 16,649 hypothesis tests, 7,775 are statistically significant at the .05 level. Among these significant effects, 15% have a different sign than the consensus UWLS estimate.[20] To see if reported estimates tend to be larger than population mean effects, we simply divide individual estimates by the corresponding UWLS estimate of the mean. In the subset of estimates that are both statistically significant and in the "correct" direction, the median estimate-to-UWLS ratio is 3.0. Thus, the significant estimates in our sample are likely to be about 3.0 times too big or wrongly signed.

These calculations are also related to the "exaggeration factor," which Ioannidis et al. (2017) calculate as $\left|\frac{\bar{\beta}-\beta^{\text{UWLS}}}{\beta^{\text{UWLS}}}\right| + 1$, where $\bar{\beta}$ is the average of estimates in a meta-analysis and

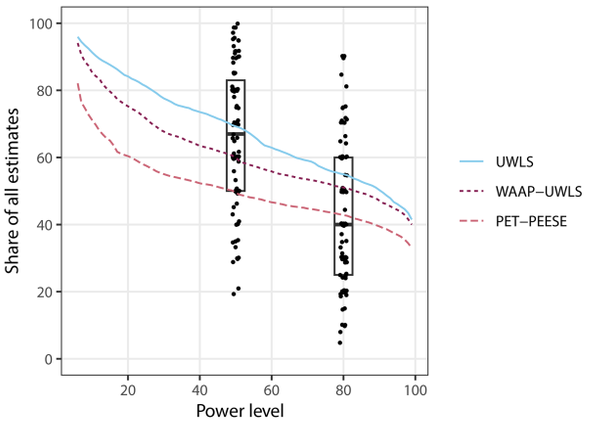

Figure 3. Share of estimates by power level, after each population mean effect estimate is inflated by a factor of 5.

β^{UWLS} is the UWLS estimate for that meta-analysis. The median exaggeration factor in our dataset is 1.9, which is close to what these authors found in economics.[21]

Sample composition

One potential objection relates to the composition of our dataset. In our expert survey, respondents stated expectations about power levels for all estimates published in well-ranked journals over the past two decades. However, our sample includes only a small share of those estimates because few studies ever get aggregated in meta-analyses (see app. A for the full list of meta-analyses in the sample), and the estimates are not all drawn from highly ranked journals.[22]

To address this concern, we asked our experts if they believed estimates in meta-analyses to be better- or worse-powered than all other estimates. In all, 75% of them think that studies included in meta-analyses (in our sample) have about the same power or higher power than other studies (out of our sample). This should make one comfortable with our comparison of expert survey estimates of power in political science and our power results drawn from meta-analyses in figure 2. This also implies that our approach of

18. If MEIs are systematically larger than meta-analytic estimates, the vast majority of political science research should also yield null results, which we do not see in fig. 1.

19. This analysis is inspired by Kollepara et al. (2021).

20. Although low power interacting with significance filtering can lead to these "errors," they do not depend in any way on how we calculated power. These results come simply from comparing the meta-analytic population mean effect to each significant coefficient within the meta-analysis.

21. Note that this exaggeration factor is the same quantity reported in Ioannidis et al. (2017) but that it is not equivalent to the type M error of Gelman and Carlin (2014). One important difference is that the exaggeration factor is computed based on the full set of estimates, whereas type M error is a characteristic of the subset of statistically significant estimates. Another difference is that unlike for type M (see fig. 2 of Gelman and Carlin [2014]), there is no simple mechanical relationship between power and the exaggeration ratio. This makes comparisons across disciplines difficult, because such comparisons will depend on rates of selective reporting, bias in the estimate of UWLS, and other factors.

22. As we noted above, results from other disciplines suggest that it is not obvious that publications in higher ranked journals should be better powered than others. Investigating this question would be an interesting path for future research.

generalizing from tests in meta-analyses to those not in meta-analyses is either accurate or overstates the power of hypothesis tests in political science in general. Appendix C reports further details about our survey and this question.

Heterogeneity across research questions

Figure 2 merges all our estimates together, but there might be variation in power levels across meta-analyses. We examine this by graphing the median power within each of the 351 meta-analyses in our dataset. Figure 4 reveals substantial heterogeneity. Although most research areas have median power below 10%, a substantial share has median power above 80%. This finding is consistent with previous assessments of power in neuroscience (Button et al. 2013; Nord et al. 2017) and economics (Ioannidis et al. 2017).[23]

One factor that may explain the high level of heterogeneity in power across study areas is that different journals may apply different evaluation standards to research. For example, if "top" journals seek to publish studies with better statistical power (which is unclear), this might account for part of the heterogeneity that we observe. Unfortunately, we do not have the data required to test this idea empirically in political science. However, prior research in economics suggests that studies in top-tier journals do not consistently exhibit higher statistical power (Askarov et al. 2024) and that highly selective venues may in fact fall prey to a "winner's curse" phenomenon (Costa-Font, McGuire, and Stanley 2013; Young, Ioannidis, and Al-Ubaydli 2008).

Power by subfield

We now turn to the question of whether power varies across subfields. To do so, we coded each meta-analytic articles in our sample into crude subfields of American politics, comparative politics, international relations, political economy, and public administration.[24] Table 1 shows the number of individual estimates, meta-analytic estimates, and meta-analysis articles we found in each subfield. The smallest number of estimates is in American politics, whereas international relations and political economy lead in terms of meta-estimates and individual estimates, respectively. Given our data collection protocol, we cannot claim that sample is balanced, but it has reasonable coverage across some of the disciplines major fields.

Figure 5 shows two ways of looking at the distribution of power by subfield. First, the dotted lines show the distribution of median power, where the median is taken for each of our 351 meta-analytic estimates. Second, the solid lines show the distribution of median power, where the median is taken at the level of our 46 meta-analytic articles. Because tests within a meta-analytic article tend to be conducted using similar data and methods, it arguably makes more sense to look at the article-based estimates (solid line) in order to avoid basing our conclusions on a few meta-analytic articles, which could have outlying results.

Overall, figure 5 shows that (a) power is low in all subfields, and (b) there is substantial variation within subfields. Of course, we have to be careful when interpreting these graphs, to avoid making an underpowered comparison about power. Nevertheless, our impressionistic sense is that the problem of lower power is not confined to any one particular subfield. Instead, there appears to be low power across the discipline.

Power over time

We examine how power has changed over time based on the publication year of each test. To reduce noise, we group each test into five-year bins from 1990 until 2020 and then calculate mean and median UWLS power per bin.[25]

Figure 6 shows that power has increased over time but remains low. Mean power rose from 16% in the first period to 28% in the last. Median power is lower and has increased less, rising from 7% to 12%. The gap between mean and median power reveals that some of the increase in mean power is being driven by increases in power at the top end of the distribution rather than a distribution-wide shift.

WHY IS POWER LOW AND WHAT CAN WE DO ABOUT IT?

Although this article has demonstrated that power is low, the sources of this problem remain less clear. Perhaps the most useful diagnostic fact is that most of the variation in power in our sample occurs across rather than within meta-analyses, with some having either very high or very low median power (see fig. 4). Differences of this size across areas of the discipline are more likely due to differences in the magnitude and stability of the effects under study rather than sample sizes. Put simply, some research areas may be studying larger and more consistent treatment effects than others. If many of the effects that political scientists study are very small, then it is not surprising to find that significant estimates often have

23. Using a somewhat different method, Szucs and Ioannidis (2017) find similarly low power in psychology, cognitive neuroscience, and medically oriented neuroscience.

24. Two coauthors independently coded the subfields, and any disagreements were resolved through a discussion that reached consensus.

25. We do not examine tests before 1990 or after 2020 as the data are sparse.

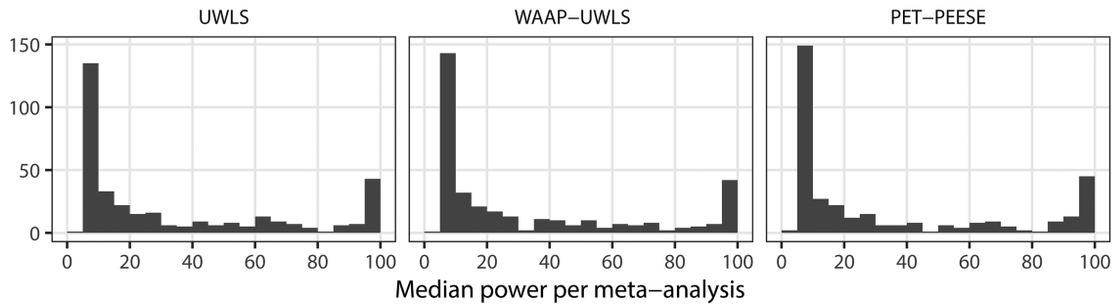

Figure 4. Histogram of median power per meta-analysis, using three different approaches to estimate the population mean effect.

the "wrong" sign and that significant and "correctly signed" effects are inflated.

At least four broad categories of interventions could be helpful in addressing these issues: increasing precision, limiting selection on significance, facilitating the evaluation of published research, and relying on theory development and qualitative methods. These interventions are not new, but they are still too rare in political science so their value bears repeating (Malhotra 2021; Nyhan 2015; Williamson et al. 2022).

First, interventions that directly increase the precision of statistical estimates can be useful. One obvious way forward would be to assemble bigger datasets, perhaps by incentivizing team-based data collection efforts. In doing so, we should be mindful that large data collection efforts are expensive, and that our quest for higher power could reinforce the discipline's existing resource inequities and hinder more speculative research agendas. In addition to collecting more data, researchers can also increase the precision of their estimates by leveraging research designs and measurement strategies that prioritize precision such as pre-post, within-subject, or repeated measures experimental designs and the use of indexes or scales combining multiple measures rather than single-item outcomes (Ansolabehere, Rodden, and Snyder 2008; Broockman, Kalla, and Sekhon 2017; Clifford, Sheagley, and Piston 2021; Hainmueller, Hopkins, and Yamamoto 2014).

Second, interventions that limit the various forms of selection on significance are important, as they ensure that low-powered studies do not introduce bias when they enter the literature. Preregistration and registered reports[26] can be very useful here (Nosek et al. 2018). Although not always possible, registered reports can limit selection on significance both during data analysis and during the publication process. More political science journals should experiment with them. More also needs to be done to reduce selection on statistical significance and the file drawer problem (Laitin 2013).[27] A necessary shift in this direction involves disciplinary norms: political science needs to do more to value null findings. Researchers need to develop the skills required to properly frame and analyze null findings (Williamson et al. 2022), using tools such as sensitivity analyses and equivalence tests (Rainey 2014).

Third, interventions that allow us to better judge the quality of already published research are important both for interpreting past work and for checking quality going forward. Data and code sharing are already common in political science (King 2003), although we could do more to share full code pipelines, from data acquisition and cleaning to analysis (Nosek et al. 2015). Less common but very useful are replications of past work, and more could be done to promote and value replications (Camerer et al. 2018).[28]

More generally, many subfields of political science still have a culture that rewards single-authored publications and lone researchers pursuing more or less standalone research programs. Moreover, editors and peer reviewers often enforce a requirement of simultaneous theoretical and empirical innovation in research articles. We worry that this

Table 1. Number of Estimates and Articles by Subfield

	Estimates	Meta Estimates	Articles
American politics	685	25	7
Comparative politics	2,150	13	11
International relations	4,438	264	5
Political economy	8,045	18	11
Public administration	1,331	31	12

26. A registered report is reviewed at the design stage, and papers are conditionally accepted before data is gathered. The *Journal of Experimental Political Science* has been accepting preregistered reports since its creation, and *The Journal of Politics* is currently piloting this approach. We note, however, that *Comparative Political Studies* piloted preregistered reports but soon abandoned this practice (Findley et al. 2016).

27. The *Political Studies Review* publishes the Null Hypothesis section, which is dedicated to research notes reporting a null finding. More journals should follow suit.

28. For instance, the Institute for Replication "works to improve the credibility of science by systematically reproducing and replicating research findings in leading academic journals." See https://i4replication.org/.

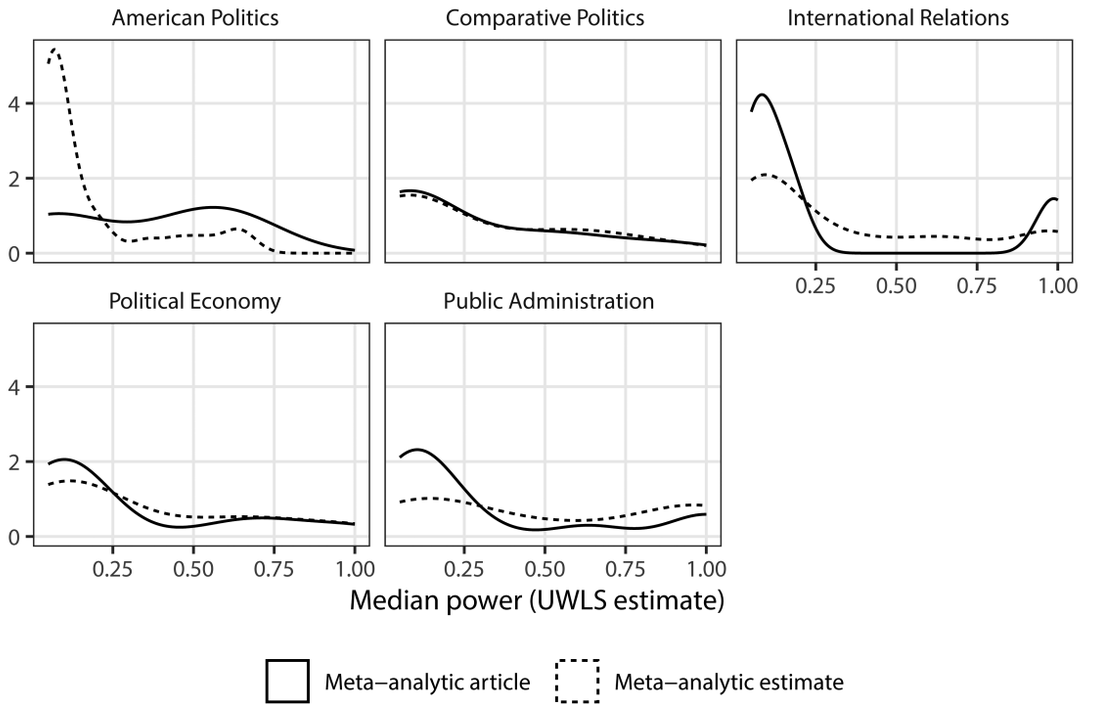

Figure 5. Densities of median power per subfield.

one-off incentive structure may lead to a proliferation of theories, which are all to be tested using single-use and often low-powered research designs. Many of the problems noted in this article would be lessened if journal editors, peer reviewers, and promotion committees did more to incentivize researchers to work in teams on shared questions tested using replicated and collaborative empirical studies.[29] We are hopeful that this would help us develop more durable and cumulative knowledge of politics.

Finally, we must accept that it is sometimes impossible to conduct an adequately powered hypothesis test. For example, researchers may simply not be able to design a well-powered study to detect small differences between a limited number of states.[30] Moreover, standard practices like estimating models with interaction terms to test conditional theories or treatment effect heterogeneity can often do more harm than good, as they can dramatically increase required sample sizes (Gelman 2018). In such cases, a useful response is to turn to theory: derive new implications and test them where data are richer. Alternatively, researchers could draw on qualitative or mixed-method approaches, using interviews, ethnographies, or archival work to shed light on causal mechanisms or heterogeneity (Gerring 2017; Small 2011). In our view, these methods are often better entry points than underpowered quantitative research designs.

CONCLUSION

Statistical power is an important and neglected aspect of quantitative political science research. This article has shown that power in political science is typically very low, that it is improving only slowly, and that this issue should be taken seriously by political scientists and methodologists. The problem of low power combined with the selection on statistical significance that occurs in our literature means that published and significant effects are likely to be much larger in print than in reality. Our best estimate is that published and statistically significant results in political science are around three times larger than the true effect under study.

Low power poses a fundamental challenge to researchers in political science. We must resist the temptation of business as usual and avoid committing the *what does not kill my statistical significance makes it stronger* fallacy (Gelman 2017).[31] Instead, our research community must address the

29. See the *Metaketa Initiative* (Dunning et al. 2019).

30. Indeed, Doucette (2024) conducts Monte Carlo simulations to study the statistical power of more than 1,100 studies on the effects of democracy using cross-national data. The author concludes that these studies generally lack the power to detect anything but very strong homogeneous effects.

31. One example of this fallacy would be to argue that a point estimate gives strong support for one's theory because it achieved statistical significance *despite* a small sample.

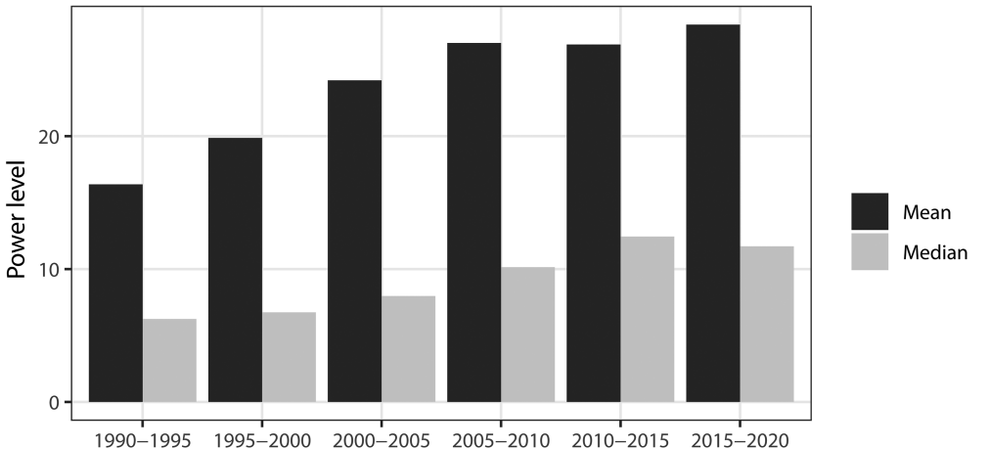

Figure 6. Power over time (UWLS).

problems of low power and selection on significance with institutional, methodological, and theoretical remedies.

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